

Improved Stabilizer Math (Complete Form)

Micro-factor: $\epsilon = 1e-7$ (0.0000001) • Fractional gain: α in (0,1), often $\alpha = 0.25$

Symbol	Meaning
x_n	input/state at cycle n
ϵ	micro-factor used to magnify small numerical drift
m_n	magnified state (division stage)
r_n	fractional margin (residual drift in magnified space)
α	fractional correction gain (how aggressively drift is removed)

1) Division (Magnify Microscopic Drift)

Amplify tiny numerical error by dividing by epsilon:

$$m_n = x_n / \epsilon$$

2) Fractional Margin Extraction

Measure drift as the residual away from the nearest stable (integer-adjacent) value:

$$r_n = m_n - \text{round}(m_n)$$

3) Fractional Correction (Stabilizer Core)

Apply a corrective force that removes only a fraction α of the residual drift:

$$c_n = -\alpha * r_n$$

4) Apply Correction in Magnified Space

Update the magnified value:

$$\begin{aligned} m_n' &= m_n + c_n \\ &= m_n - \alpha * r_n \end{aligned}$$

5) Restore Scale (Multiply Back)

Return to original magnitude:

$$x_{\{n+1\}} = m_n' * \epsilon$$

Closed-Form Update Equation

Substituting steps 1-5 gives the full stabilizer equation in one line:

$$x_{\{n+1\}} = \epsilon * (x_n/\epsilon - \alpha * (x_n/\epsilon - \text{round}(x_n/\epsilon)))$$

Compact Form

$$x_{\{n+1\}} = x_n - \alpha * \epsilon * (x_n/\epsilon - \text{round}(x_n/\epsilon))$$

Feedback and Convergence

Because x_{n+1} feeds back as the next input, the residual (fractional margin) contracts geometrically. A standard approximation for the residual dynamics is:

$$\begin{aligned} r_{n+1} &\approx (1 - \alpha) * r_n \\ r_n &= r_0 * (1 - \alpha)^n \end{aligned}$$

Example (alpha = 0.25)

Each cycle removes 25% of remaining residual drift in magnified space:

$$r_n = r_0 * (0.75)^n$$

Interpretation: epsilon provides a numerical microscope (division), the fractional margin isolates drift (residual), alpha controls correction strength, and the multiply-back returns the system to the original scale each cycle.